Faculty of Computer Science and Engineering
Hochiminh City University of Technology

Kỹ thuật dữ liệu
Data Engineering
Course ID: CO5173

Semester: 261

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Course objective

- Analyzing and designing databases using different data models, including relational and non-relational models
- Database indexing, transaction processing, flexible data processing, and query optimization
- Knowledge and skills related to data warehousing, big data, and data management architectures for modern applications
- Advanced research topics in data engineering and emerging applications.

Course learning outcomes

1. L.O.1. Design databases that meet the requirements of database applications
    1. L.O.1.1. Design databases at different levels: conceptual, logical, and physical
    2. L.O.1.2. Model data using different data models, ranging from relational to non-relational models
2. L.O.2. Design data warehouses and data management architectures that integrate data from various sources to support users' decision-making needs
    1. L.O.2.1. Design data management architectures that integrate data from different data sources)
    2. L.O.2.2. Design data warehouses: Understand concepts and fundamental techniques in data warehouse, data mining, BI, database security
3. L.O.3. Manage and process data in modern application environments
    1. L.O.3.1. Manage complex data: Understand NoSQL, BigData techniques
    2. L.O.3.2. Process data using modern data processing paradigms
    3. Be able to conduct research on modern databases and applications

Course Outline

PART A: Lecturer

- **W1 (05/09/26):** Introduction to the course and lecture methods
  Review techniques in Database design (ER Model)
- **W2 (12/09): ER Weakness + EER model**
  Relational database model, ER/EER to relational model mapping
- **W3 (22/9): Data retrieve using SQL**
  <Practice activities: Database designing tools and DBMS>

PART B: Student presentation (30% of the total score)

- **W3 (19/09)**
  - **Database Design Theory & Methodology: FD and Normalization (G1)**
  - **Data storage and indexing introduction/fundamental (G2)**
- **W4 (26/09)**
  - **Advanced indexing techniques: B/B+ tree other from research papers or used in modern DBMS (G3)**
  - **Physical DB design, Database tuning in relational systems + Give examples, case studies,… (G4)**

Course Outline

- **W5 (03/10):**
  - **Big data concepts and applications: NoSQL Vs. Relational DB, MongoDB (G5)**
  - **Data warehouse (DW) and Decision support systems (G6)**
- **W6 (10/10):**
  - **FD & Normalization (G7)**
  - **Database security (G7)**
  - **Data mining, BI tools and Demonstration (G8)**

(17/10): Off

- **W7 (24/10):**
  - **Represent a research paper in DE (G9)**
  - **R/R* tree and other from research papers or used in modern DBMS (G10)**
  - **Review/exercise**
  - **Prepare for Mini-project presentation**

Course Outline

Part C: Project presentation: 30% (W8-10: 31/10, 07/11, 14/11)

- Each session: 4 + 3 + 3 groups
- 15 minute presentation + 20 minute Q&A
- Every member must present
- Member's contribution must be clearly claimed

Evaluation

- 30%: Presentation (absence -> total score = 0)
- 30%: Project (absence -> total score = 0)
- 40% (Final Exam): Multi-choice questions

Note:

- To pass the course: accumulated point >= 5.5
- **absent 2 lectures: -1 point from total score, -1 more score for each further absence**
- W1: decide the presentation chapter and the project topic for each group (student)
- Submit slides 2 days prior the presentation date to:

https://drive.google.com/drive/folders/1lOKpzO5Q8MnaTerd1DbwRoKdABbeO5ip?usp=sharing

Project Topics (recommended)

<Can use tools from Oracle, MS SQL Server Integration Services & Analysis Services, Pentaho,…)

1. Design a data warehouse + BI for a banking system
2. Design a data warehouse + BI for commercial system
3. Investigate MS Azure to develop an DW/BI system
4. Investigate data warehouse and mining techniques in Google cloud platform (GCP)
5. Query Optimization methods/techniques
6. NoSQL case studies
7. Investigate BigData techniques: Hadoop, Spark, Hive
8. Search techniques in modern database systems, eg. ElasticSearch, Cassandra
9. Investigate data lake, lake house techniques

Project Topics (recommended)

10. Data warehouse and ELT techniques
11. SQL for data analysis
12. Python for Data analysis
13. R for Data analysis
14. Traffic state estimation using data mining techniques (e.g., prediction, classification, clustering, association rule mining)
15. Data crawling techniques, e.g., to crawl data about traffic condition in HCMC
16. Routing optimization considering traffic condition
17. Optimizing data organization and access in large databases
18. Data management strategies and techniques in enterprises

Project Topics (recommended)

19. Database systems for IoT
20. Database for mobile, light-weight systems
21. Investigate about TensorFlow and its applications
22. Securities in database
23. Read and re-present papers about advanced data techniques from top conferences/journals:
    - **SIGKDD : ACM SIGKDD International Conference on Knowledge discovery and data mining**
    - **VLDB : International Conference on Very Large Databases**
    - **SIGMOD : ACM SIGMOD International Conference on Management of Data**
      ICDE - International Conference on Data Engineering
    - IEEE Transactions on Knowledge and Data Engineering (**TKDE**)

Reference

[1] R. Elmasri, S.B. Navathe: *"Fundamentals of Database Systems"*, 7th Edition, Pearson Addison-Wesley, 2016
[2] R. Kimball, M. Ross, "The Data Warehouse ToolKit", 3rd Edition, Wiley Publishing, Inc., 2013.
[3] Trần Minh Quang, et al. "Cơ sở dữ liệu: Khái niệm, phương pháp và kỹ thuật phân tích thiết kế", NXB ĐHQG HCM (tái bản) 2025 .
[4] J. Hurwitz, A. Nugent, F. Halper, M. Kaufman: "*Big Data for Dummies*", John Wiley & Sons Inc., 2013
[5] Jiawei Han, Micheline Kamber, Jian Pei, "Data Mining: Concepts and Techniques", Third Edition, Morgan Kaufmann Publishers, 2012.
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[7] Jure Leskovec, Anand Rajaraman, Jeffrey D. Ullman: Mining of Massive Datasets, 2014.
[8] S. Lightstone, T. Teorey, T. Nadeau: "*Physical Database Design*", Elsevier Inc., 2007

Q&A

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